

Name: _____

NetID: _____

ECON 1001 — Fall 2025 — Final Exam

Version A

There are 23 multiple choice questions and 2 free response questions. Select the best answer using the models from the course. Each MC problem and each FR subproblem is worth 4 points.

1. Assume that allowing a cheat sheet does not affect the exam or how it is graded. Also assume that the benefits and costs of having a cheat sheet could be measured in dollars for each student. Allowing a cheat sheet on the exam would be efficient
 - A) ♦ If, for most students, the benefits of the cheat sheet outweigh the costs of the cheat sheet.
 - B) ■ If the sum of the total benefits each student gets from a cheat sheet were greater than the sum of the total costs to each student.
 - C) • If, before the exam, most students voted that a cheat sheet should be allowed.
 - D) Statements ♦ and ■ are both correct.

2. A new government imposes a tax system where people pay 20% on income from \$0–\$20,000; 15% on income from \$20,000–\$40,000; and 10% on income above \$40,000. (For example, if someone has an income of \$25,000, they pay 20 percent on the first \$20,000 and 15 percent on the next \$5,000.)
 - A) ♦ People with higher annual incomes pay more taxes (in dollars).
 - B) ■ The tax system is regressive, because higher-income people have a lower average tax rate than people with lower incomes.
 - C) • The tax system is progressive, because higher-income people have a higher marginal tax rate than people with lower incomes.
 - D) Statements ♦ and ■ are both correct.

3. Diminishing marginal utility of income
 - A) Implies risk aversion.
 - B) Justifies redistributing money from high-income people to low-income people on utilitarian grounds. (Recall that a utilitarian aims to maximize the sum of individual utilities of each person in society.)
 - C) Does *not* mean that very rich people don't care about more money.
 - D) All of the other options are correct.

4. A profit-maximizing firm that faces a flat demand curve
 - A) Has no market power.
 - B) Has an MR curve beneath its demand curve.
 - C) Does not necessarily sell at the quantity where $MR = MC$.
 - D) Sells a differentiated product.

5. A firm with market power
 - A) ♦ Sets the price as high as it thinks it can while still maintaining some sales.
 - B) ■ Maximizes average profit per unit.
 - C) • Sells the quantity where $MR = MC$.
 - D) Statements ■ and • are both correct.

6. For a firm with market power (that cannot price discriminate), marginal revenue is
 - A) Equal to the price.
 - B) The money obtained from the last customer.
 - C) Equal to average revenue.
 - D) Less than the price, because of the discount effect.

7. Assuming the product already exists, if a firm has market power, then relative to setting the price where $MC = AR$,
 - A) Producer surplus is higher.
 - B) Consumer surplus is lower.
 - C) Deadweight loss is larger.
 - D) All of the other options are correct.

8. Governments tend to supply (or heavily regulate) electricity because
 - A) Otherwise firms would have little incentive to provide it.
 - B) It is an important source of government revenue.
 - C) The government is especially good at executing complicated projects.
 - D) A free market for electricity would tend to be very inefficient.

9. Suppose for a firm there were no fixed costs and average variable costs were constant. Then
 - A) The AC curve is U-shaped.
 - B) The AC curve is flat.
 - C) The firm will struggle to make profits.
 - D) The price would fall to the average fixed cost.

10. In Figure 1 (last page)
 - A) The firm should sell the quantity that maximizes the gap between AC and demand.
 - B) We can determine the profit-maximizing price even though no units are displayed on the axes.
 - C) Profit is zero at the quantity-price pairs where AC and demand intersect.
 - D) None of the other options is correct.

11. Suppose all firms in a market sell an identical good and have the same cost curves. Entry will occur
 - A) As long as there is no lobbying by the operating firms.
 - B) Until all firms earn zero economic profit.
 - C) Until each firm provides exactly one unit of the good.
 - D) None of the other options is correct.

12. If a firm can perfectly price discriminate,
- A) Some consumers are better off than if it cannot.
 - B) Deadweight loss is zero (assuming no externalities).
 - C) There is no need to advertise.
 - D) None of the above options is correct.
13. A college sells tickets to two football games: the **Big Game** (against a big rival) and the **Big Win** (against a very weak team). There are two groups of fans (A and B), each with 1,000 people. The college can sell 2,000 tickets per game (assume $MC = 0$). Table 1 shows the WTP for each type of fan.
- A) Charge 5 for the Big Game and 3 for the Big Win.
 - B) Charge 4 for the Big Game and 2 for the Big Win.
 - C) Charge 7 for a bundle of two tickets (one for each game).
 - D) Charge 6 for a bundle of two tickets (one for each game).
14. Suppose Mr. X and Mr. Y face the same purely financial bet. Mr. X takes it; Mr. Y does not. Assuming both are risk-averse, a possible explanation is
- A) Mr. X is richer.
 - B) Mr. X's utility curve has a different shape from Mr. Y's.
 - C) The bet is not a fair bet.
 - D) All other options are possible explanations.
15. Policy A increases total economic surplus by \$10 million. Policy B increases the surplus by only \$5 million and is less equitable than Policy A. The mayor chooses Policy B. Then we know
- A) ♦ The mayor is corrupt.
 - B) ■ The mayor does not care about efficiency at all.
 - C) • The mayor cannot redistribute surplus however she likes.
 - D) Statements ♦ and ■ are both correct.
16. Talia's initial utility is 13. She considers a bet: with 60% probability she loses \$1,000 (her utility would fall to 10), and with 40% probability she wins \$4,000 (her utility would rise to 20). Will she take the bet?
- A) Yes.
 - B) No.
 - C) It depends on her initial wealth.
 - D) She is indifferent.
17. Advertising
- A) Increases efficiency.
 - B) Reduces efficiency.
 - C) Shifts a firm's demand curve inward toward the vertical axis.
 - D) Can increase **or** decrease efficiency.

18. A movie theater had been naively charging \$10 for all tickets. After an (accurate) analysis, you determine it should charge \$12 for non-students and \$8 for students. If adopted:
- A) Producer surplus will definitely increase.
 - B) Consumer surplus will definitely increase.
 - C) Total economic surplus will definitely increase.
 - D) All of the other options.
19. Devan's Donuts uses a super-secret ingredient. The cost of that ingredient rises by \$1 per donut. Assume straight-line demand and MR curves and a flat MC curve.
- A) Devan raises the price by exactly \$1.
 - B) Devan raises the price by more than \$1.
 - C) Devan raises the price, but by less than \$1.
 - D) Devan keeps the price the same.
20. Malte sells cupcakes. He paid his annual rent (a fixed cost) in January. By July he realizes total costs for this year will exceed total revenue for this year. Moreover, he knows if he stays in business his economic profit next year will also be negative. In July, Malte
- A) Stops selling immediately.
 - B) Might stop immediately or might decide to continue to the year's end.
 - C) Continues to sell this year but not next year.
 - D) Raises the price.
21. Consider the demand in Table 2.
- A) ♦ When lowering price from 7 to 6 or from 5 to 7, the output effect dominates the discount effect.
 - B) ■ For some quantities MR is positive and for others it is negative.
 - C) • The firm might optimize at a price of 5.
 - D) Statements ■ and • are both correct.
22. Two competitors agree to actions that raise profits. They simultaneously choose whether to **Honor** or **Break** the agreement. Payoffs are in Table 3. What happens?
- A) A Honors, B Honors.
 - B) A Breaks, B Breaks.
 - C) A Breaks, B Honors.
 - D) A Honors, B Breaks.
23. Now suppose the same game is **sequential**, with A moving first. What happens?
- A) A Honors, B Honors.
 - B) A Breaks, B Breaks.
 - C) A Breaks, B Honors.
 - D) A Honors, B Breaks.

Respond to problems 24 and 25 on the back side of your response sheet.

24) Consider the story of an incumbent firm and barriers to entry. As always, assume all the individuals are rational and purely self-interested. **Answer only on your response sheet.**

- A professor for a class assigns a Classic Textbook to his 1,000 students.
- The author of the Classic Textbook earns \$1 profit per student.
- Morris has written a New Textbook which is cheaper and just as good as the Classic Textbook. Morris does not require any payment for the book.
- If the professor assigned the New Textbook, each student would benefit \$2 relative to having to use the Classic Textbook.
- The professor is very easy to influence (by providing him nice meals and gifts).

- a) What is the efficient outcome? Briefly explain on your response sheet.
- b) Why is it likely the professor will *not* assign the new book? Briefly explain on your response sheet.
- c) Under what conditions would the professor assign the New Textbook? Continue to assume that all individuals are rational and self-interested. Briefly explain on your response sheet.

25) Based on the information in the table, answer the following questions. **Answer on the back of your Response Sheet (do not answer on this page).**

$Q = 10 - P$
$MR = 10 - 2Q$
$MC = 6$
$AC = 6 + \frac{2}{Q}$

- a) What is the *efficient* quantity?
- b) What is the profit-maximizing price?
- c) What is the DWL?
- d) What is the profit?

DO NOT TURN OVER UNTIL INSTRUCTED TO.

Write your name and Canvas ID *on both sides* of your response sheet. We will only grade the response sheet.

Good luck!

Figure 1

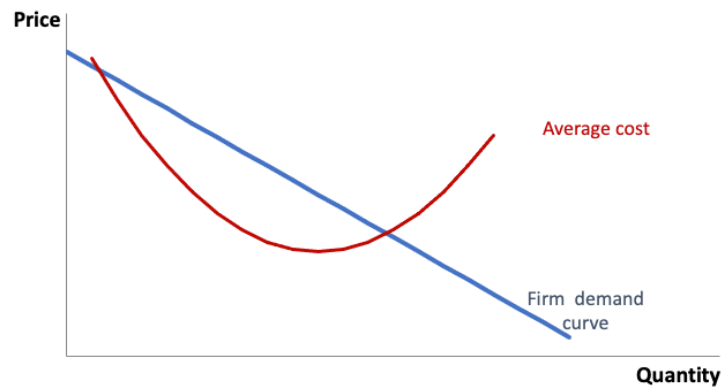


Table 1

	Group A's WTP	Group B's WTP
Big Game	5	4
Big Win	2	3

Table 2

P	Q
7	5
6	6
5	7

Table 3

	B chooses to Honor	B chooses to Break
A chooses to Honor	(6, 6)	(4, 7)
A chooses to Break	(7, 4)	(5, 5)

Note: Payoffs are (A's payoff, B's payoff).